The examination is being carried out on the **following application documents**

Description, Pages

1-8, 10-18	as published	
9	filed in electronic form on	24-06-2019

Claims, Numbers

1-12	filed in electronic form on	07-11-2018

Drawings, Sheets

1/2, 2/2	as published

1 Clarity (Art. 84 EPC):

1.1 Despite the deletion of the very confusing passage of the description in page 9, lines 5 to 7, the definition given to the wording "microfibrillated cellulose" is still very broad: the non delimiting terms such as "normally" and "often" used in page 9, lines 3 to 4 should be deleted. Furthermore, the wording "the microfibrils may also be longer, for example between 10 to 200 micrometers " in page 9, line 5, is much broader than what is conventionally considered as a typical length range of microfibrillated-cellulose which is normally only of several micrometers. In the document ISO/TS20477:2017 introduced by the applicant, the length is typically up to 100 micrometers. The fact that the dimensions of the MFC are so broad combined with the absence in the description of a specific method for obtaining said MFC (the wording mechanically delamination is once again very vague and an homogenization step is only mentioned in the examples) renders the definition of the MFC unclear in the present application.

The cited passage in page 9, line 5 that "the microfibrils may also be longer, for example between 10 to 200 micrometers" should therefore also be deleted.

1.2 After a careful reconsideration of the application, it is not clear how to distinguish the MFC prepared from carboxymethylated fibers, MFC prepared from oxidised fibers and/or the MFC prepared by CMC-functionalized fibers (as a water retention agent) from the microfibrillated cellulose MFC in claims 1 and 11: finally, these different forms of MFC remains MFC and it is unclear if they should be counted as the MFC or as the water retention agent in claims 1 and 11.

2 Cited documents:

D1 SYVERUD, K. ET AL.: "Strength and barrier properties of MFC films";", CELLULOSE,
vol. 16, no. 1, 2009, pages 75-85, XP008153597,

D17 EP 2 557 225 A1 (IMERYS MINERALS LTD [GB]) 13 February 2013 (2013-02-13)

D18 Davide Beneventi ET AL: "Highly Porous Paper Loading with Microfibrillated Cellulose by Spray Coating on Wet Substrates", INDUSTRIAL & ENGINEERING CHEMISTRY RESEARCH,
vol. 53, no. 27, 9 July 2014 (2014-07-09), pages 10982-10989, XP055573733,
ISSN: 0888-5885, DOI: 10.1021/ie500955x

3 Lack of inventive step (Art. 56 EPC)

The arguments made by the Applicant have been carefully reviewed, however a lack of inventive step is upheld in view of D1 and D17.

The problem-solution approach argued in our last communication dated 29703/2019 is maintained. Paragraph [0097] of D17 clearly contradict the applicant's assertion that D17 is not concerned with the creation of barrier coating layers. Furthermore, D17 clearly and unambiguously discloses that CMC, polyacrylates, gums, alginates, acrylic thickeners are known water retention aids or thickeners which can be used in the coating composition comprising microfibrillated cellulose (paragraphs [0106]-[0107]). The amount of CMC as an additive added in the coating composition in example 22 is within the scope of the present claims 1 and 11. Besides, the examiner disagrees that

paragraph [0111] of D17 excludes coating on wet substrates: in this paragraph, on-machine coating , i.e. coating on the sheet forming machine is disclosed too. The examiner agrees with the applicant that the coating composition of D17 further comprises fillers and latex in example 22, however the wording "the coating composition comprises..." used in claims 1 and 11 of the present application does not exclude the presence of these components either.

Starting from D1 which discloses a wet in wet coating of MFC comprising composition on a paper substrate, the skilled person would be prompted in the light of D17 to add the CMC as a thickening agent or a water retention aid in the coating composition found in D1, since D17 teaches that the presence of this CMC contributes to a high level of water retention and rheology control to the coating formulation. These characteristics are important in any paper coating step, whether it is for a coating on wet or on a dry substrate.

Claims 1 and 11 thus lack an inventive step in view of D1 and D17 (Art. 56 EPC).

In a similar way, claims 1 and 11 lack an inventive step in view of D18 combined with the teaching of D17 (Art. 56 EPC): as a matter of fact, D18 discloses that spray coating of MFC slurries in an amount of above 6 g/m^2 on wet papers allow the complete retention of MFC on the surface of the substrate along with a negligible MFC loss during the dewatering stage (paragraph "conclusions2 in page 10987). To further improve the water retention and rheology control of the MFC coating formulation, the skilled person would turn to D17 and arrive at the subject-matter of claims 1 and 11 in an obvious way.

It is finally to be noted that, in the present application, the improvement of the barrier resistance has only been showed for CMC as a water retention agent in an amount of 3 to 8.5 wt% based on the weight of MFC (see table 2). It seems that the the claimed technical effect is not obtained for all the components listed in claim 1 and 11.

4 Further prosecution

The applicant is invited to file a new independent claim which takes account of the above comments. The applicant should also indicate how the subject-matter of the new claim differs from the state of the art and the significance thereof.

In order to comply with the requirements of Rule 137(4) EPC, the applicant should clearly identify the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the passages of the application as filed on which these amendments are based (see Guidelines E-II, 1).